Analytical Chemistry Cumulative Exam
March 8, 2013

Method validation is an important aspect of analytical chemistry. This cumulative exam tests your understanding of the basics of method validation.

1. (15 points) Analytical chemists often report their results in terms of confidence intervals such as the 95% confidence interval. Explain the following using words or figures.
   a) What is a confidence interval?
   b) How are confidence intervals derived for small data sets?
   c) Why has 95% confidence been chosen as the common level for confidence intervals for reporting analytical chemistry derived data?

2. (10 points) Provide an analytical chemistry definition of precision. How is precision related to accuracy?

3. (15 points) Provide a graphical illustration of LOD. Using this graph, comment on the assumption that at the LOD the analyst knows with 99% confidence that an instrumental signal at or greater than the LOD is due to the analyte.

4. (10 points) What is an interference? What methods during an analysis can be used to overcome interferences?

5. (10 points) What is an internal standard? Why are internal standards used in some methods?

6. (15 points) Briefly explain how curve fitting is performed. You may use any fit in your example such as a linear least squares fit or a 5-parameter logistic fit.

7. (15 points) Residual plots from fit calibration curves such as the least squares fit can give you significant information about the quality of the fit.
   a) What is a residual plot? Use illustrations if it helps justify your answer.
   b) Why is using a residual plot better than using correlation coefficient, r^2, for determining appropriateness of a curve fit?

8. (10 points) Comment on the following statement. Instrumental methods of analysis are preferred over classical methods of analysis such as titrations and gravimetric analyses because instrumental analyses are more accurate and precise as compared to classical methods of analysis.